

Detección de Anomalías Acústicas en Maquinaria Industrial

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Motivación



Motivación



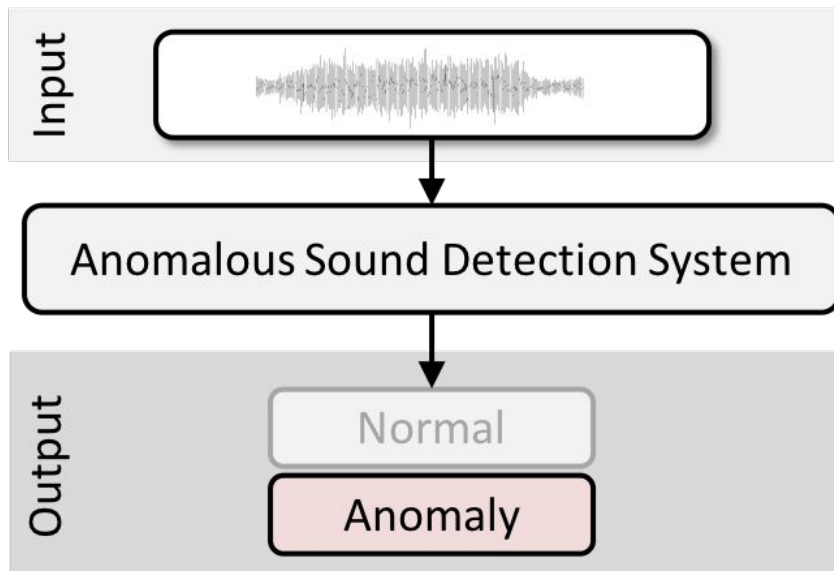
- Tiempos de inactividad no planificados

Motivación

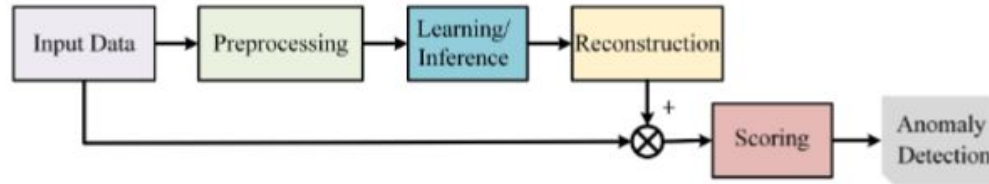


- Tiempos de inactividad no planificados
- Mantenimiento reactivo

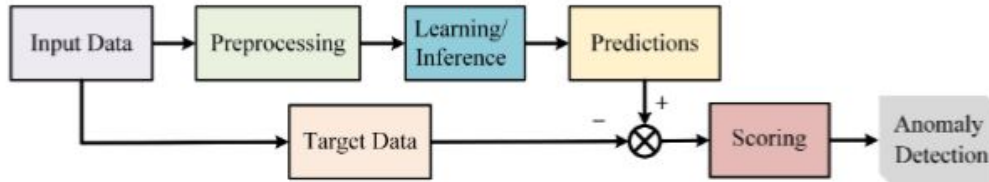
Alcance



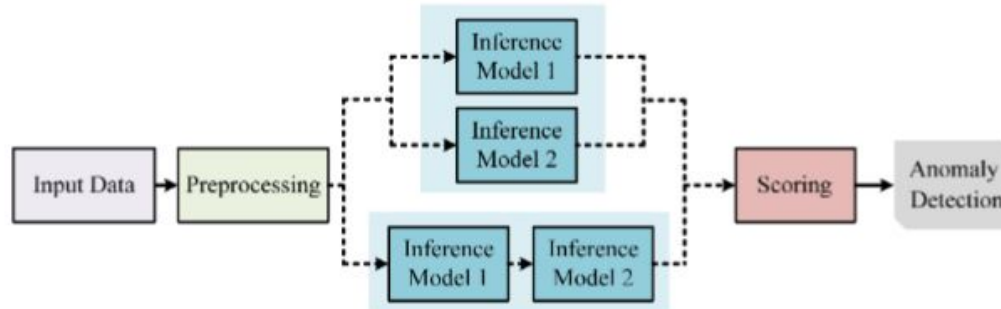
Detectar anomalías acústicas en maquinaria industrial mediante el análisis de espectrogramas de audio, utilizando técnicas de deep learning



(a) Reconstruction-based Approach



(b) Prediction-based Approach



(c) Hybrid Method

First-Shot Unsupervised Anomalous Sound Detection for Machine Condition Monitoring



The aim of this task is to develop anomalous sound detection techniques that can train models on new data with noisy normal machine sounds and a few additional samples containing only factory noise or clean normal machine sounds, enabling the model to achieve higher detection performance regardless of environmental noise shifts or other domain shifts.

Organizers



Tomoya Nishida
Hitachi, Ltd.



Daisuke Niizumi
NTT, Inc.



Roberto Sannino
STMicroelectronics



Filippo Augusti
STMicroelectronics



Kota Dohi
Hitachi, Ltd.



Takashi Endo
Hitachi, Ltd.



Noboru Harada
NTT, Inc.



Davide Albertini
STMicroelectronics



Simone Pradolini
STMicroelectronics



Keisuke Imoto
Kyoto University



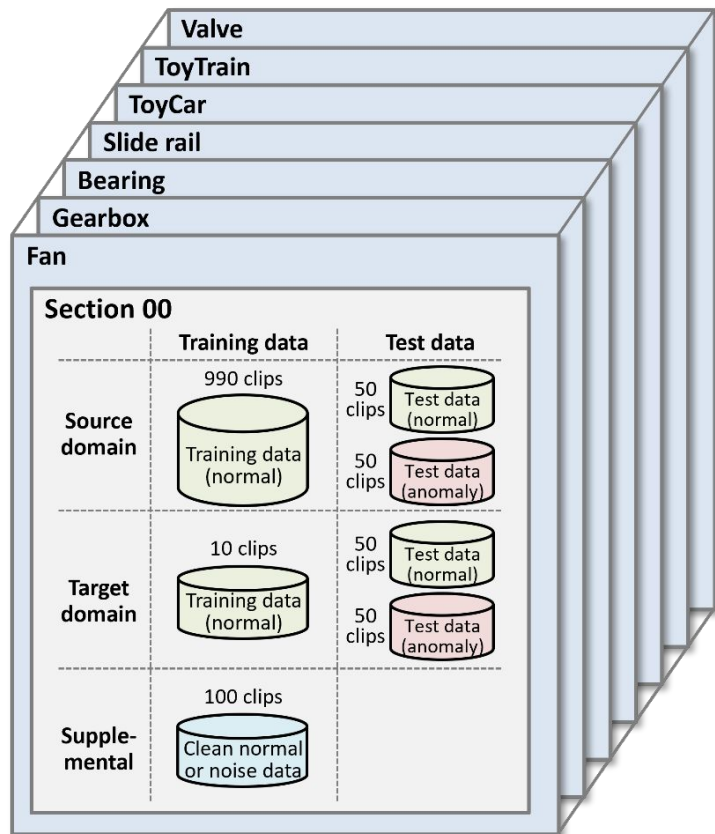
Harsh Purohit
Hitachi, Ltd.



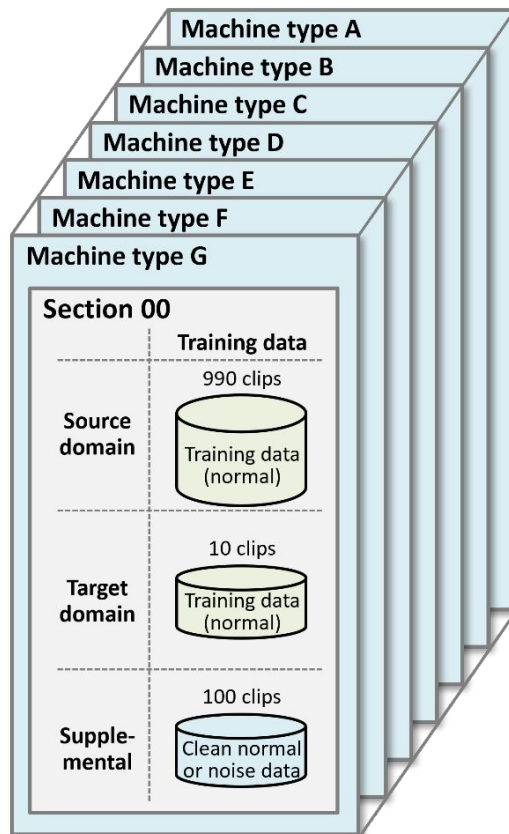
Yohei Kawaguchi
Hitachi, Ltd.



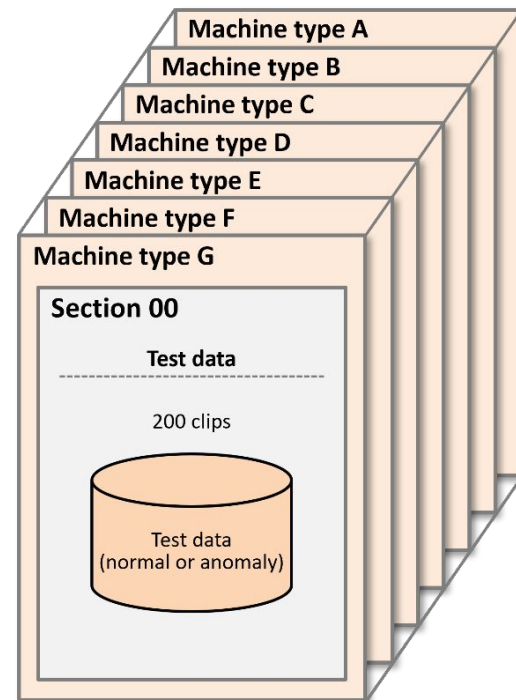
Development dataset



Additional training dataset



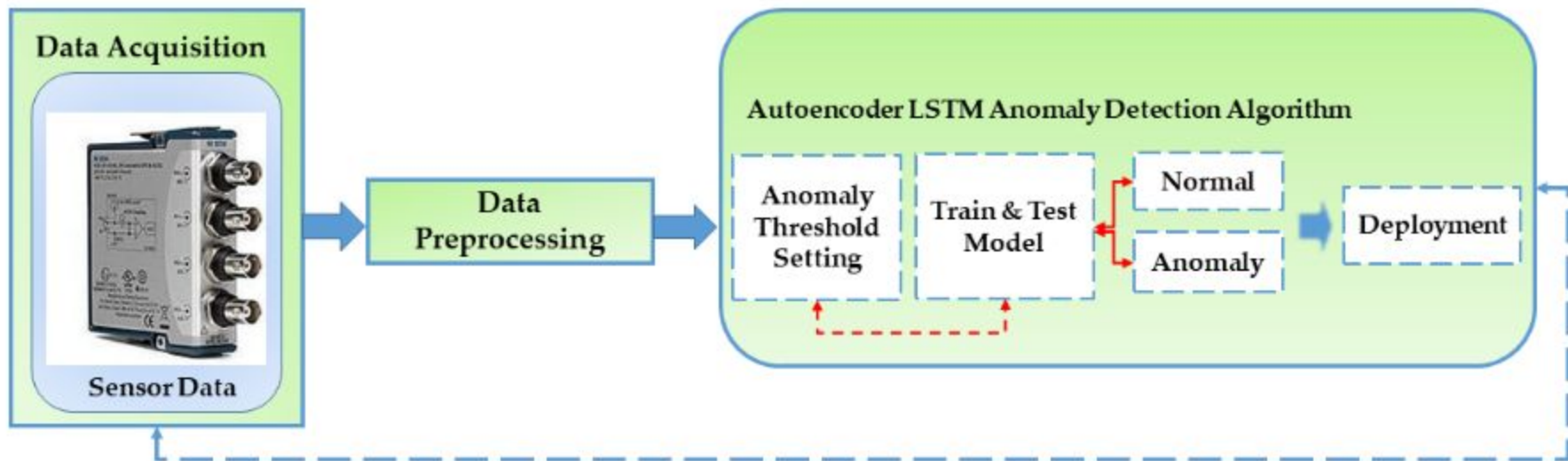
Evaluation dataset

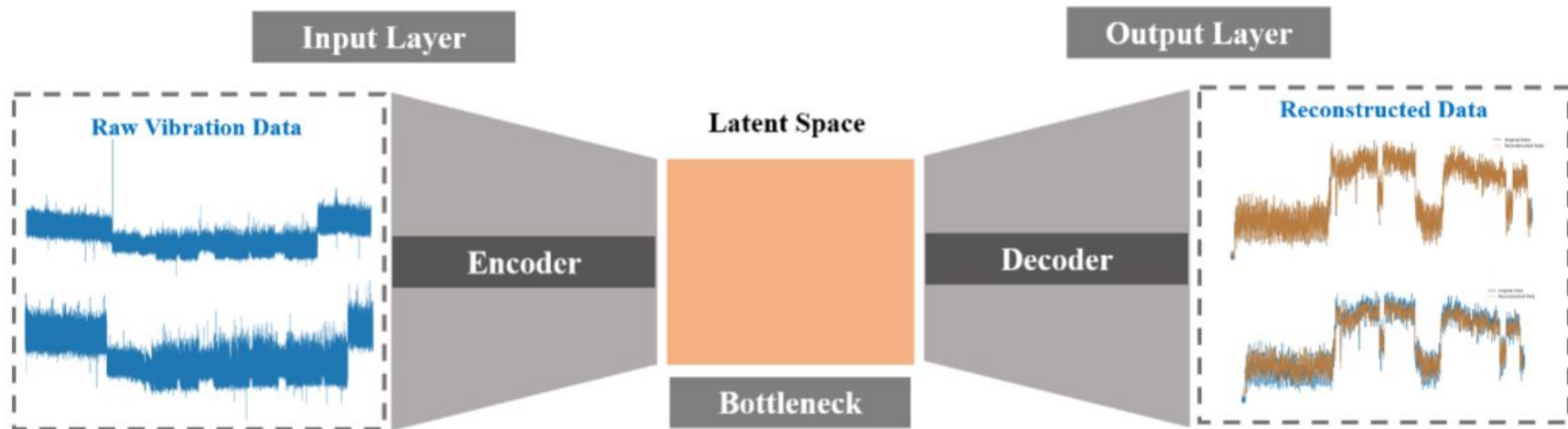


Datos

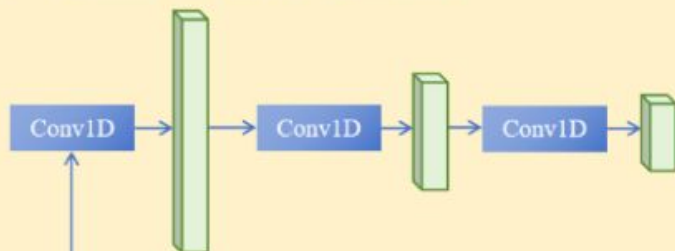
- Sonidos de operación normal y anómala de 7 tipos de máquinas reales/de juguete:
 - ToyCar, ToyTrain, Fan, Gearbox, Bearing, Slide rail, Valve
- Monocanal, 10-12 segundos por clip, incluye sonido de la máquina + ruido ambiental
- 990 clips normales (dominio source) + 10 clips normales (dominio target) + 100 clips suplementarios (sonidos limpios normales o solo ruido)

Estado del Arte

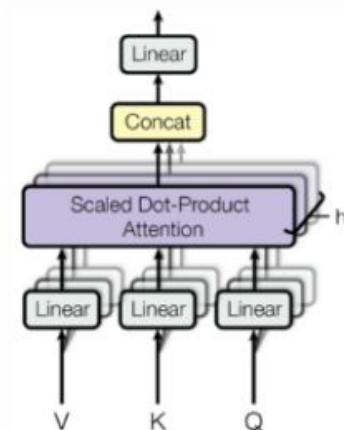




Convolutional Feature Extraction



Multi-Head Attention Mechanism



LCAA

